Chapter – 1

Introduction

A way to measure human progress is through inventions that ease mental and physical burdens. The digital computer is one such great invention.

Digital computers have made possible many scientific, industrial and commercial advances that would have been unattainable otherwise.

Characteristics of a digital system is its manipulation of discrete elements are electric impulses, decimal digits, letters of an alphabet, arithmetic operations, punctuation marks, any other meaning full symbols.

Digital Computer (Discrete information processing system)

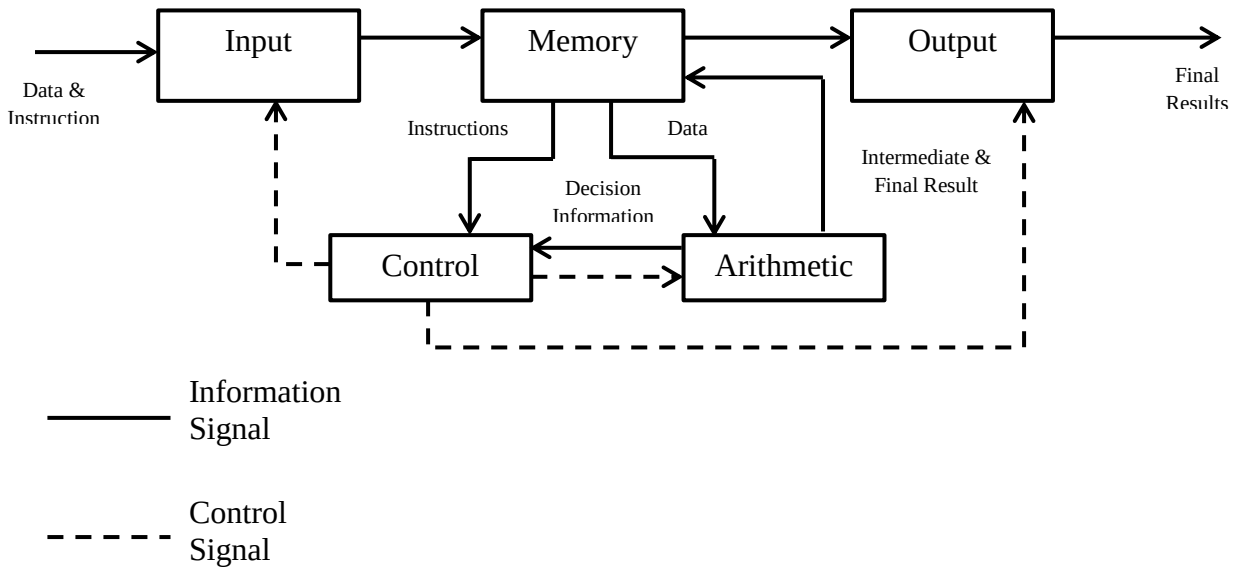


Fig1: Digital Computer

A digital computer is an interconnection of digital modules. Above diagram showed that a digital computer manipulates discrete elements of information and that these elements are represented in the binary form. Operands used for the calculations may be expressed in the binary number system; even decimal digits are represented in binary codes.

Data processing is carried out by means of binary logic elements using binary signals. Quantities are stored in binary storage elements.

Binary Numbers

Decimal to Binary Conversions

Ex:1 $(7392)_{10} \rightarrow (1110011100000)_2$

$\Rightarrow 7 \times 10^3 + 3 \times 10^2 + 9 \times 10^1 + 2 \times 10^0$

Note: [In general, a number with a decimal point is represented by a series of coefficients hence the power of 10 by which the coefficient must be multiplied - - - - $a_5\, a_4\, a_3\, a_2\, a_1\, a_0\, a_{-1}\, a_{-2}$- - - - - - $10^5 a_5 + 10^4 a_4 + 10^3 a_3 + 10^2 a_2 + 10^1 a_1 + 10^0 a_0 + 10^{-1} a_{-1} + 10^{-2} a_{-2}$].

2	7392	
2	3696	----0
2	1848	----0
2	924	----0
2	462	----0
2	231	----0
2	115	----1
2	57	----1
2	28	----1
2	14	----0
2	7	----0
2	3	----1
	1	----1

= **1110011100000**

Ex:2 $(0.6875)_{10} \rightarrow (0.1101)_2$

0.6875 x 2 = 1.3750 1

0.3750 x 2 = 0.7500 0

0.75 x 2 = 1.5000 1

0.5 x 2 = 1.0000 1

Binary to Decimal Conversion

Ex:1 11010.11 → 26.17

$= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 + 1 \times 2^{-1} + 1 \times 2^{-2}$

= 16 + 8 + 0 + 2 + 0 + 0.50 + 0.25

= **26.75**

Decimal to Octal Conversions

Ex:1 $(0.513)_{10} \rightarrow (0.40651767)_8$

0.513 x 8 = 4.104	4
0.104 x 8 = 0.832	0
0.832 x 8 = 6.656	6
0.656 x 8 = 5.248	5
0.248 x 8 = 1.984	1
0.984 x 8 = 7.872	7
0.872 x 8 = 6.976	6
0.976 x 8 = 7.808	7

Octal and Hexadecimal numbers Conversions

Ex:1Binary to Octal

$(1\,0\,1\,1\,0\,0\,0\,1\,1\,0\,1\,0\,1\,1\,.\,1\,1\,1\,1\,0\,0\,0\,0\,0\,1\,1\,0)_2 \rightarrow (26153.7406)_8$

$(\ 2\ \ 6\ \ 1\ \ 5\ \ 3\ .\ 7\ \ 4\ \ 0\ \ 6\ )_8$

Ex:2 Binary to Hexadecimal

$(1\,0\,1\,1\,0\,0\,0\,1\,1\,0\,1\,0\,1\,1\,.\,1\,1\,1\,1\,0\,0\,0\,0\,0\,1\,1\,0)_2 \rightarrow (2C6B.F06)_{16}$

$(\ 2C\ \ 6\ \ B.\ \ F\ \ 06\ )_{16}$

Complements

Complements are used in digital computers for simplifying the subtraction operation and for logical manipulations.

There are two types of complements.

1) r's Complement.

2) (r-1)'s Complement.

Complement for Decimal Numbers

10's Complement

Ex:1 $(52520)_{10} \rightarrow 47480$ [Note: Number of digits in the number is n=5]

$= 10^5 - 52520$

= 1,00,000 – 52520

= **47480**

Ex:2 $(0.3267)_{10}$

$= 10^0 - 0.3267$

= 1 – 0.3267

= **0.6733**

9's Complement

Ex:1 $(52520)_{10} \rightarrow 47479$

$= (1 - (10^5 - 52520))$ **OR** $(10^5 - 10^0 - 52520)$

$= (1 - 47480)$ $(1,00,000 - 1 - 52520)$

= **47479** **47479**

Ex:2 $(0.3267)_{10} \rightarrow 0.6732$

$= (1 - 10^{-4} - 0.3267)$

$= (0.9999 - 0.3267)$

= **0.6732**

Complement for Binary Numbers

2's Complement

Ex:1 101100 → 010100

No. of digits n=6

$2^6 = 64$

2	64	
2	32	----0
2	16	----0
2	8	----0
2	4	----0
2	2	----0
	1	----0

$(1000000 - 101100)_2$

1000000

–101100

0010100

⇒ **010100**

1's Complement

Ex:1 101100 → 010011

No. of digits n=6

$2^6 = 64$ $(2^6 - 1) = 63$

2	63	
2	31	----1
2	15	----1
2	7	----1
2	3	----1
	1	----1

$(111111 - 101100)_2$

111111

101100

010011

⇒ **010011**

2's Complement

Ex:2 $(0.0110)_2 \rightarrow 0.1010$

n=0

$= 2^0 = 1$

$= (1 - 0.0110)$

= **0.1010**

1's Complement

Ex:2 $(0.0110)_2$

$= (2^0 - 2^{-4} - (0.0110))$

$= ((1 - 0.0625) - 0.0110)$

$= ((0.9375)_{10} - 0.0110)$

$= (0.1111 - 0.0110)$

0.1111

(–) 0.0110

0.1001

0.937 x 2 = 1.875 1

0.875 x 2 = 1.750 1

0.750 x 2 = 1.500 1

0.500 x 2 = 1.000 1

Subtraction with r's Complements

The subtraction of two positive numbers (M – N), both of base r, may be done as follows:

1. Add the minuend M to the r's complement of the subtrahend N.
2. Inspect the result obtained in step1 for an end carry,
   a. If an end carry occurs, discard it.
   b. If an end carry does not occur, take the r's complement of the number obtained in step1 and place a negative sign in front.

Ex:1 Subtract (75322 – 3250) using 10's complement.

M = 75322
N = 03250
10's complement of N
$\Rightarrow$ n = 5, $10^n = 10^5 = 1,00,000$
$\Rightarrow (10^5 - 3250)$
$\Rightarrow$ 96750

```
      75322
(+)   96750
      171072
```

Answer is**71072**

Ex:2 Subtract $(3250 - 75325)_{10}$ using 10's complement.

M = 03250
N = 75325
10's of N
75325
$10^5 = 1,00,000$
(100000 – 75325)
= 24675

```
      03250
(+)   24675
      27925
```

No carry so take 10's complement and
Place a negative sign.
Answer is $(10^5 - 27925)$
–72075

Ex:3 M = 1010100
N = 1000100
2's complement of N
1000100 $\Rightarrow$ 128

```
2 | 128
2 |  64  ----0
2 |  32  ----0
2 |  16  ----0
2 |   8  ----0
2 |   4  ----0
2 |   2  ----0
      1  ----0
```

```
      10000000
      01000100
      00111100
        1010100
(+)     0111100
        0010000
```

Answer is **0010000**

Subtraction of M,N numbers using (r-1)'s complement

Ex:1 M = 72532
N = 03250

9's complement of N = 03250
$(10^5 - 1 - 3250)$

96749
M = 72532
+ N = 96749
169281
1
69282

Answer is **69282**

Ex:1 M = 1010100
N = 1000100

Using 2's complement
2's complement of N
1's complement 0111011
2's complement 0111011
+ 1
0111100

1010100
+ 0111100
End Carry → 10010000

Answer is **10000**

Ex:2 M = 03250
N = 72532

9's complement of 72532
$(10^5 - 1 - 72532)$

27467
M = 03250
+ N = 27467
30717

→ No carry, so take 9's complement
$= -(10^5 - 1 - 30717)$
= **– 69282**

Ex:2 M = 1000100
N = 1010100

Using 2's complement of N
1's complement 0101011
2's complement 0101011
+ 1
0101100

1000100
+ 0101100

No end carry so take 2's complement of answer and put negative sign.
1' complement → 0001111
+1
0010000 Answer is **–10000**

Basic Theorems and Properties of Boolean algebra

1.	$x + 0 = x$	$x . 1 = x$
2.	$x + x' = 1$	$x . x' = 0$
3.	$x + x = x$	$x . x = x$
4.	$x + 1 = x$	$x . 0 = 0$
5.	$(x')' = x$ [Involution]	
6.	$x + y = y + x$	$x . y = y .x$ [Commutative]
7.	$x + (y + z) = (x + y) + z$	$x(yz) = (xy)z$ [Associative]
8.	$x(y + z) = xy + xz$	$x + yz = (x + y)(x + z)$ [Distributive]
9.	$(x + y)' = x' y'$	$(x y)' = x' + y'$ [Demorgan]
10.	$x + x y = x$	$x(x + y) = x$ [Duality]

Minterms and Maxterms

Minterms → (Standard Product)

A binary variable may appear in its normal form (x) or in its normal form (x) or in its complement form (x′), combination of those variables with an AND operation are called Minterms.

Ex: $(x' y' z' + x y' z + x' y z + x y z' + x y z)$ [$m_{[]}$]

Maxterms →

A binary variable may appear in its normal form or in its complement form combination of those variables with a OR operation are called Maxterms.

Ex: $(x + y + z)(x' + y + z')(x + y'+ z')(x' + y' + z)(x' + y' + z')$ [$M_{[]}$]

Sum of minterms

For n binary variables, one can obtain 2^n distinct minterms. Any Boolean function can be expressed as a sum of minterms.

Ex:1 Express the Boolean function $F = A + B'C$ in a sum of minterms.

Step 1: Expressing A in F as sum of minterms

$$A = A(B + B')$$
$$= AB + AB'$$

Still missing on more variable C

$$A = AB + AB'$$
$$= AB(C + C') + AB'(C + C')$$
$$= ABC + ABC' + AB'C + AB'C'$$

Step 2: Expressing second term B′C as sum of minterms.

$$B'C = B'C(A + A')$$
$$= AB'C + A'B'C$$

Step 3: Combining all terms

$$F = A + B'C$$
$$= ABC + ABC' + AB'C + AB'C' + AB'C + A'B'C$$

$x + x = x$

$$= ABC + ABC' + AB'C + AB'C' + A'B'C$$

Reaming the minterms in ascending order,

$$F = A'B'C + AB'C' + AB'C + ABC' + ABC$$
$$= m1 + m4 + m5 + m6 + m7$$
$$F(A, B, C) = \Sigma(1, 4, 5, 6, 7)$$

[Note: Σ this denotes ORing of terms]

Product of Maxterms

Each of 2^n function of n binary variables can be also expressed as a product of Maxterms.

Ex:1 Express the Boolean function F = xy + x′z

Step 1: Conversion of function F into OR terms using distributive law.

$$F = xy + x'z$$
$$= (xy + x')(xy + z)$$
$$= (x + x')(y + x')(x + z)(y + z)$$
$$= (x' + y)(y + z)(x + z)$$

Step 2: Expressing OR terms with missing variable.

$$(x' + y) = x' + y + zz' = (x' + y + z)(x' + y + z')$$
$$(x + z) = x + z + yy' = (x + z + y)(x + z + y')$$
$$(y + z) = y + z + xx' = (y + z + x)(y + z + x')$$

Step 3: Combining all terms

$$F = (x' + y + z)(x' + y + z')(x + z + y)(x + z + y')(y + z + x)(y + z + x')$$

$x . x = x$

$$= (x' + y + z)(x' + y + z')(x + z + y')(x + y + z)(y + z + x')$$

$x . x = x$

$$= (x + y + z)(x + z + y')(x' + y + z')(y + z + x')$$

$= M_0 M_2 M_5 M_4$

$F(x, y, z) = \Pi(0, 2, 5, 4)$

[Note: Π this denotes ANDing of Maxterms]

Simplification of Boolean functions

An implementation of a Boolean function requires the use of logic gates. A smaller number of gates, with each gate having less number of inputs, may reduce the cost of the implementation.

There are 2 methods for simplification of Boolean functions.

1. The algebraic method by using Identities.
2. The graphical methods by using Karnaugh map method.

Karnaugh Map Method

The map is a diagram made up of square. Each square represents one minterms. Since any Boolean function can be expressed as a sum of minterms. The map presents a visual diagram of all possible ways a function may be ex pressed in a standard form.

By recognizing various patterns, the user can derive alternative algebraic expressions for the same function, from which select the simplest one.

[Note: The simplest algebraic expression is any one in a sum of products or product of sums that has a minimum number of literals.]

Literal: A variable or its complement.

Two and three variable maps

- There are four minterms for two variables; hence the map consists of four squares, one for each minterms.

m_0	m_1
m_2	m_3

x \ y	0	1
0	$x' y'$	$x' y$
x { 1	$x' y'$	$x y$

- The 0's and 1's marked for each row and each column designate the values of variables x & y respectively.

Note: X appears primed in row 0 and unprimed in row 1. Similarly, y appears primed in column 0 and unprimed in column 1.

Ex:1 Simplify the Boolean function.

$x'y + xy' + xy$

$\Rightarrow$ $x'y + xy' + xy = m_1 + m_2 + m_3$

x \ y	0	1
0		1
1	1	1

$\Rightarrow$ $x + y$

If we mark the squares whose minterms being to a given function, the two variable map becomes another useful way to represent any one of the 16 Boolean functions of two variables.

- There are eight minterms for three binary variables.

m0	m1	m3	m2
m4	m5	m7	m6

x \ yz	00	01	11	10
0	$x'y'z'$	$x'y'z$	$x'yz$	$x'yz'$
1	$x\,y'z'$	$x\,y'z$	xyz	xyz'

Note: Any to adjacent squares in the map differing by only one variable which is primed in one square and unprimed in the other.

Ex:2 Simplify the Boolean function.

$F = x'yz + x'yz' + x\,y'z' + x\,y'z$

x \ yz	00	01	11	10
0			1	1
1	1	1		

$F = x'y + x\,y'$

Ex:3 Simplify the Boolean function

F = x′yz + xy′z′ + xyz + xyz′

x \ yz	00	01	11	10
0			1	
1	1		1	1

I

II

F = yz + xz′

Ex: 4 Simplify the Boolean function

F = A′C + A′B + AB′C + BC

A \ BC	00	01	11	10
0		1	1	1
1		1	1	

I

II

F = C + A′B

Ex: 5 Simplify the Boolean function

F(x,y,z) = ∑(0, 2, 4, 5, 6)

x \ yz	00	01	11	10
0	1			1
1	1	1		1

I

II

F = z′ + xy′

Ex: 6 Simplify the Boolean function

F(x,y,z) = ∑(2, 3, 6, 7)

x \ yz	00	01	11	10
0			1	1
1			1	1

I

F = y

Ex: 7 Simplify the Boolean function

$F(A,B,C) = \sum(7, 3, 4, 5)$

A \ BC	00	01	11	10
0			1	
1	1	1	1	

$F = BC + AB'$

Ex: 8 Simplify the Boolean function

$F(A,B,C,D) = \sum(4, 6, 7, 3)$

AB \ CD	00	01	11	10
0			1	
1	1		1	1

$F = CD + AD'$

Ex:9 An automobile alarm circuit is used to detect certain undesirable conditions. There are three switches used to indicate the status of the door adjacent to the driver's seat, the ignition and the headlights respectively. Design the logic circuit with these three switches as the inputs, so that the alarm will be activated whenever either of the following conditions exists.

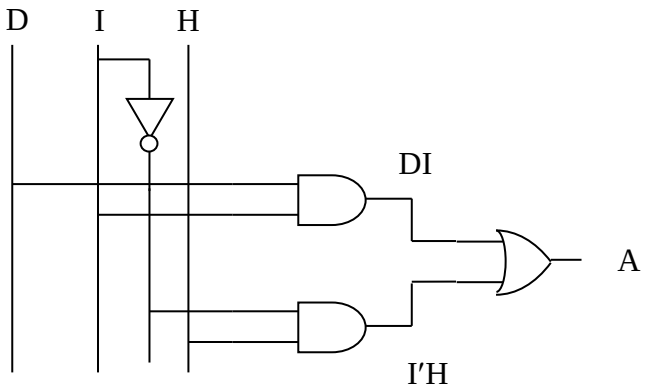
a) The head lights are ON while the ignition is OFF.

b) The door is open while the ignition is ON.

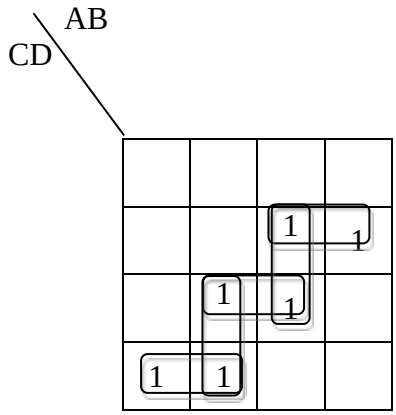
D	I	H	A
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

D \ IH	00	01	11	10
0		1		
1		1	1	1

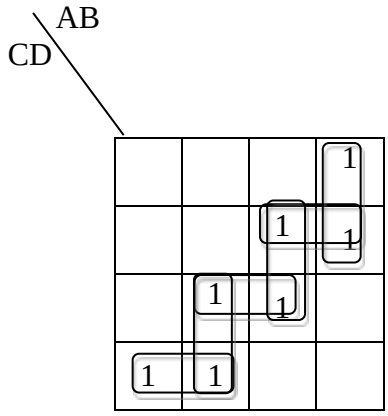
$A = I'H + DI$



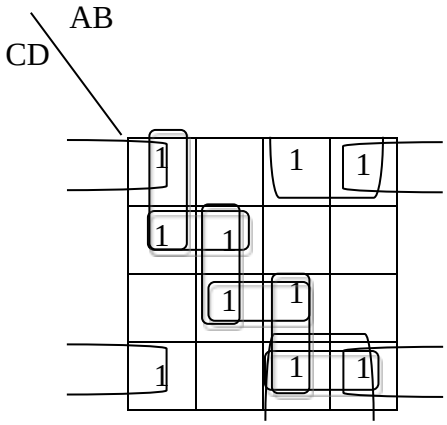
1.



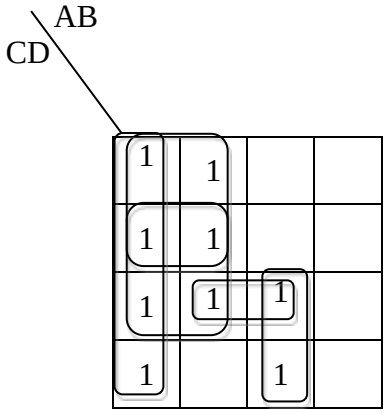
3.



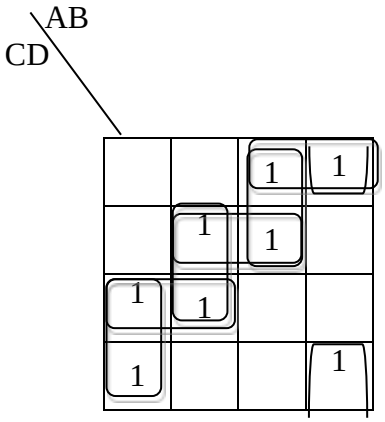
2.



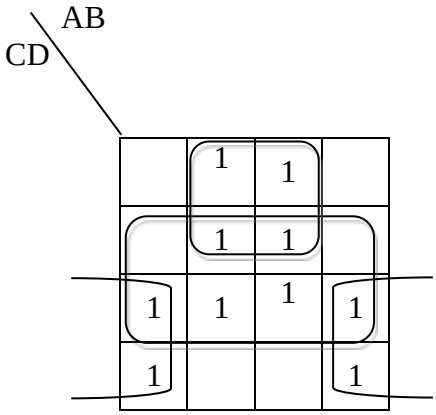
4.



5.



6.



Essential Prime Implicants

These prime Implicants that cover an output of the function that no combination of other prime Implicants is able to cover.

5 – Variable K-Map

n = 5

$2^5 = 32$

F(A,B,C,D,E) =

AB \ CDE	000	001	011	010	110	111	101	100
00	0	1	3	2	6	7	5	4
01	8	9	11	10	14	15	13	12
11	24	25	27	26	30	31	29	28
10	16	17	19	18	22	23	21	20

Obtain the minimal SOP for the given function.

F(A, B, C, D, E) = ∑ m(9, 11, 15, 13, 25, 27, 31, 29)

AB \ CDE	000	001	011	010	110	111	101	100
00								
01		1	1			1	1	
11		1	1			1	1	
10								

⇒ BE

A	B	C	D	E
0	1	0	0	1
0	1	0	1	1
1	1	0	0	1
1	1	0	1	1
0	1	1	1	1
0	1	1	0	1
1	1	1	1	1
1	1	1	0	1

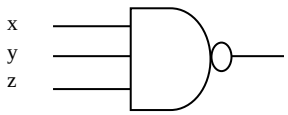
6 – Variable K-Map

n = 6

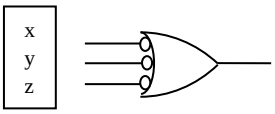
$2^6 = 64$

F(A, B, C, D, E, F) =

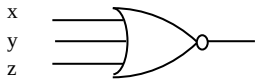
ABC \ DEF	000	001	011	010	110	111	101	100
000	0	1	3	2	6	7	5	4
001	8	9	11	10	14	15	13	12
011	24	25	27	26	30	31	29	28
010	16	17	19	18	22	23	21	20
110	48	49	51	50	54	55	53	52
111	56	57	59	58	62	63	61	60
101	40	41	43	42	46	47	45	44
100	32	33	35	34	38	39	37	36



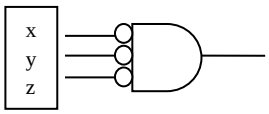
F= (x y z)′



F=x′+y′+z′ = (x y z)′

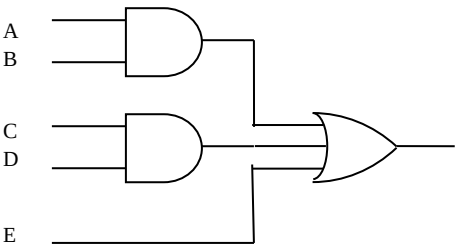


F= (x+y+z)′

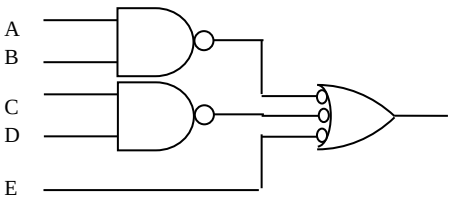


F =(x′y′z′) = (x+y+z)′

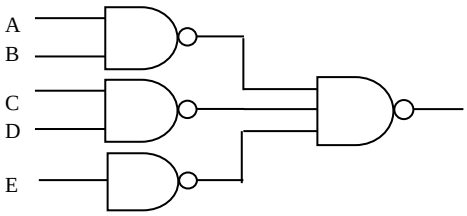
Example: F = AB + CD + E



F



F



F

Don’t Case terms consideration in K-Map

The combinations are usually obtained from a truth table that lists the conditions under which the function is a ‘1’. The function is assumed equal to ‘0’ under all other conditions.

This assumption is not always true since these are applications where certain combinations of input variables never occur.

Ex: (BCD to Excess-3 system) after decimal digit 9, (10, 11, 12, 13, 14, 15)other combinations are all in valid.

These combinations of input variables never occurred are called Don’t care terms. These can be used on a map to provide further simplification of theof the function. To distinguish the don’t care conditions from 1’s and 0’s, an ‘x’ will be used.

Example : Simplify the Boolean function F(w,x,y,z) = ∑(1, 3, 7, 11, 15) + ∑d(0, 2, 5)

wx \ yz	00	01	11	10
00	x	1	1	x
01		x	1	
11			1	
10			1	

Combining 1’s & x’s

F = yz + w′z

wx \ yz	00	01	11	10
00	x			x
01	0	x		0
11	0	0		0
10	0	0		0

Combining 0’s & x’s

F = z′ (w′ + y)

Unit 1 Question Bank

1. Realize the function F using NAND gates only,

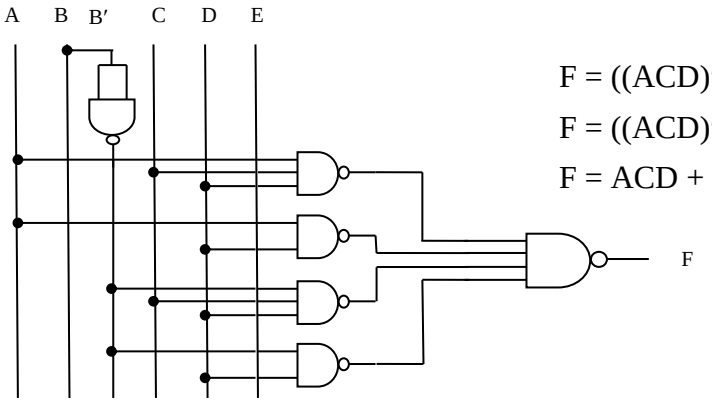
$F = (A + B') (CD + E)$

Ans. If we convert POS to SOP we can design function using NAND gates only.

$F = (A + B') (CD + E)$

Using distributive law,

$F = [ACD + AE + B'CD + B'E]$



$F = ((ACD)'.(AE)'.(B'CD)'.(B'E))$

$F = ((ACD)'' + (AE)'' + (B'CD)'' + (B'E)'')$

$F = ACD + AE + B'CD + B'C$

2. Identify the essential prime Implicantsin the function below

$F = \sum m(0, 2, 4, 7, 8, 10, 11, 12, 14, 15)$

AB \ CD	00	01	11	10
00	1			1
01	1		1	
11	1		1	1
10	1		1	1

Essential prime Implicants

$\Rightarrow$ AC, A′B′C′, C′D′, BCD

3. Which are the adjacent cell numbers of all '8' in a four variable K-Map.

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

Adjacent cell numbers of cell 8 are cell 0, cell 10 cell 9, cell 12

4. Prove that the sum of all the minterms of a Boolean function of 3 variable in one.

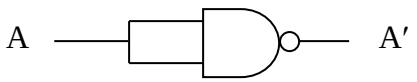
$$F = A'B'C' + A'B'C + A'BC' + A'BC + AB'C' + AB'C + ABC' + ABC$$

Rearranging the terms.

$$\begin{aligned} F &= ABC + ABC' + AB'C + AB'C' + A'BC + A'BC' + A'B'C + A'B'C' \\ &= AB(C + C') + AB'(C + C') + A'B(C + C') + A'B'(C + C') \\ &= AB + AB' + A'B + A'B' \\ &= A(B + B') + A'(B + B') \\ &= A + A' \\ &= 1 \end{aligned}$$

5. Show that logic diagram of realizing an inverter using a 2 input NAND gate.

Ans.



6. The maxterm corresponding to decimal 10 is

Ans. Binary equivalent of 10 → 1010 considering 4 variables ABCD to indicate 1010;

Maxterm⇒ $(A' + B + C' + D)$

(POS form)

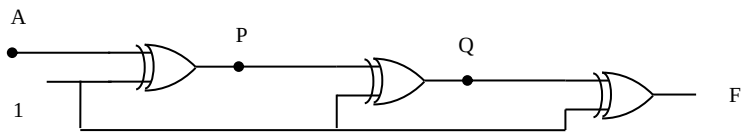
7. $(AFE.94)_{16}$ = (?)

1010 1111 1110 . 1001 0100

5 3 76 . 2 2 4

$(AFE.94)_{16} = (5376.224)_8$

8.



Ans.

Output at P = A ⊕ 1

= A′.1 + A.1′

= A′ + 0 = A′

Output at Q = A′⊕ 1

= A′′.1 + 1′.A′

= A + 0 = A

Output at F = A ⊕ 1

= A′.1 + A.1′

= A′ + 0 = A′

9. Find the minimum SOP for that function f(a, b, c) = M_2 . M_4 using K – Map. Ans.

a \ bc	00	01	11	10
0	1	1	1	0
1	0	1	1	1

F = (a′b′) + (ab) + (b′c) + (bc) + (ac)

10. Obtain the POS expression for the SOP expression is

x \ yz	00	01	11	10
00	1	1	1	0
01	0	1	0	0
11	0	0	0	0
10	1	1	1	0

F(A B C D) = B′D′ + B′C′ + A′C′D

F = (C′ + D′) (A′ + B′) (B′ + D)

F is in POS form.

11. $(CAFÉ.94)_{16} = (\ ?\ )_8$

Ans. $\underbrace{1}_{1}\underbrace{1\,0\,0}_{4}\underbrace{1\,0\,1}_{5}\underbrace{0\ \ 1\,1}_{3}\underbrace{1\,1\ \ 1}_{7}\underbrace{1\,1\,0}_{6}\,.\,\underbrace{1\,0}_{2}\underbrace{0\,1\ \ 0}_{2}\underbrace{1\,0\,0}_{4}$

$(145376.224)_8$

12. Given that $16_{10} = 100_b$, the value of b = <u>(2)</u>

13. In a 5 – Variable K – Map cell 3 is adjacent to _____ & cell 30 to _____.

AB \ CD	000	001	011	010	110	111	101	100
00	0	1	3	2	6	7	5	4
01	8	9	11	10	14	15	13	12
11	24	25	27	26	30	31	29	28
10	16	17	19	18	22	23	21	20

Adjacent to cell 3 → 1, 2, 7, 11, 19

Adjacent to cell 30 → 14, 22, 26, 28, 31

14. The minimum number of universal gates required to implement the POS expression.

Ans. $Y' = (A + B')(A' + B)$

$Y = A'B + AB'$

$Y = A \oplus B$

Using 4 NAND gate we can construct 1 XOR gate.

15. $(4321.23)_{(5)}$ = _____ $_{(10)}$

Ans. $= 4 \times 5^3 + 3 \times 5^2 + 2 \times 5^1 + 1 \times 5^0 + 2 \times 5^{-1} + 3 \times 5^{-2}$

$= 500 + 75 + 10 + 1 + 0.4 + 0.12$

$= 586.52_{(10)}$

16. $(D00D)_{(16)}$ = _____$_{(8)}$

Ans. $\underbrace{1}_{1}\underbrace{1\,0\,1}_{5}\underbrace{0\,0\,0}_{0}\underbrace{0\ \ 0\,0}_{0}\underbrace{0\,0\ \ 1}_{1}\underbrace{1\,0\,1}_{5}$

$(150015)_{(8)} = (D00D)_{(16)}$

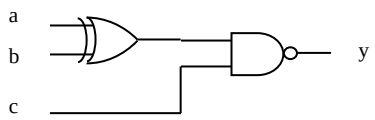
17. A XNOR 1

$A \otimes 1 = A'.1' + A.1$

$= A'.0 + A.1$

$= 0 + A = A$

18. The minimal Boolean expression for the output of the logic circuit shown in Fig 1 is



$y = ((a \oplus b).c)'$

$= (a \oplus b)' + c'$

$= a'b' + ab + c'$

19. A combinational circuit has inputs A, B, C and its K – Map is shown in Fig 2. The output of the circuit is given by,

x \ yz	00	01	11	10
0		1		1
1	1		1	

Ans. $F = A'B'C' + A'BC' + AB'C' + ABC$

20. For the identity $AB + A'C + BC = AB + A'C$, the duel form is

$AB + A'C + BC = AB + A'C$

$(A' + B')(A + C')(B' + C') = (A' + B')(A + C')$

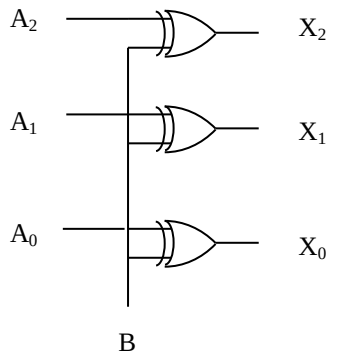
21. The maxterm corresponding to decimal number 15 is _____

Binary equivalent of 15 → 1111

Considering the variables ABCD to indicate 1111,

Maxterm → $(A' + B' + C' + D')$

22. In Fig 1 X_2, X_1 X_0 will be 1's complement of A_2, A_1 A_0 if _____.



When B = 0

X_2, X_1 X_0 outputs become complement to A_2, A_1 A_0 inputs.

23. A full adder can be implemented with half adders and OR gates. A 4-bit parallel full adder without any initial carry requires ___8___ half adder and _____ OR gates.

For one full adder

$S = A \oplus B \oplus Cin$

$C = (AB) + Cin(A \oplus B$

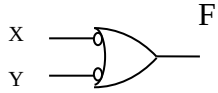
24. In a certain application 4 input's A, B, C, D are fed to logic circuit, producing an output which operates a relay. The relay turns ON when if (A, B, C, D) = 1 for the following states of the input 0000, 0010, 0101, 0110, 1101 & 1110. State 1000, 1001 do not occur and for the remaining states the relay is OFF. The minimized Boolean expression is,

A	B	C	D	R
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	X
1	0	0	1	X
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

AB \ CD	00	01	11	10
00	1			1
01		1		1
11		1		1
10	X	X		

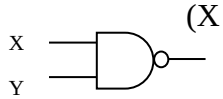
$R \Rightarrow A'B'C' + BC'D + BCD'$

25. A NOR gate with two active low inputs is equivalent to gate 2 input NAND gate.



$F = X' + Y'$

$= (X\,Y)'$



$(X\,Y)'$

26. Realize the function $F = (A + B')\,(CD + E)$ using NAND gates only

$F = (A + B')\,(CD + E)$

$= ACD + AE + B'CD + B'E$

27. Write the dual of function $F = AB + A'C + BC + AB + A'C$

$F = (A' + B')\,(A + C')\,(B' + C')\,(A' + B')\,(A + C')$

28. Identity essential prime implicates in the function given below.

$F = \sum m(0, 2, 4, 7, 8, 10, 11, 12, 14, 15)$

E.P.I 1) $C'D'$ 3) $A'B'C'$

2) AC 4) BCD

AB \ CD	00	01	11	10
00	1			1
01	1		1	
11	1		1	1
10	1		1	1

29. For a digital system with 3 inputs and Z is an output. The output should be high when atleast 2 input are high. The minimal expression for the output is,

ABC	Z
000	0
001	0
010	0
011	1
100	0
101	1
110	1
111	1

A \ BC	00	01	11	10
0			1	
1		1	1	1

$\boxed{F = BC + AC + AB}$

30. In a full adder, sum expression has __4__ essential prime implicants and carry expression has __3__ essential prime implicants.

Sum = ∑m (1, 2, 4, 7) Carry = ∑m (3, 5, 6, 7)

A \ BC	00	01	11	10
00		1		1
01	1		1	

All are EPI

A \ BC	00	01	11	10
00			1	
01		1	1	1

All are EPI

31. The complement of the Boolean expression AB(C + AC) is _____.

32. The logical expression Y = A′ + AB is equivalent to A′ + B.

Y = A′.A′ + A B ⇒ (A′ + A) (A′ + B) (A′ + B)

33. Develop the truth table & write a switching expression for the following statement.

X is a0 only if A, B, C are all 1's or if only one of the variable is a1.

A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

A \ BC	00	01	11	10
00		0		0
01	0		0	

X = (A + B + C′) (A + B′ + C)

= (A′ + B + C) (A′ + B′ + C′)

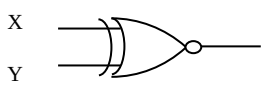
34. A = AB gets simplified to __A(A + B)__.

A.A + A.B

(A + A) (A + B)

A(A + B)

35. The POS form of XNOR operation is



XY′ + YX′ ⇒ (X′ + Y) (Y′ + X)

36. The min number of logic gates needed to realize the logic expression

Y = ABC′ + AB′C′ + AB′C + ABC is

A \ BC	00	01	11	10
0				
1	1	1	1	1

F = A

37. The canonical form of the logical expression

Y = A′C + AB is

38. The complement of the Boolean expression Y = AB (C + AC).

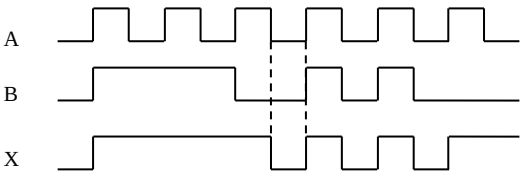
39. Convert the following SOP to an equivalent POS

ABC + A′B′C′ + ABC′ + A′B′C

40. Apply Demorgan's theorem to the expression.

(X(YZ′ + YZ))′

41. What is the logic circuit that accepts A,B as input & X as output.



1. Simplify the Boolean expression, F and show its implementation using NAND gates.

F(A, B, C, D, E) = Σm (0, 1, 4, 5, 16, 17, 21, 25, 29)

AB \ CDE	000	001	011	010	110	111	101	100
00	1	1					1	1
01								
11		1					1	
10	1	1					1	

F = A′B′D′ + AD′E + AB′C′D′

2. Simplify the Boolean expression, F and show its implementations using NAND gates:

F(A, B, C, D, E) = A′ B′ C′ E′ + B′ C′ D′ E′ + A′ B′ D′ + B′ CD + A′CD + A′BD

AB \ CDE	000	001	011	010	110	111	101	100
00	1	1			1	1	1	1
01			1	1	1	1		
11								
10					1	1		

F = A′ B′ C′ D′ + AB′CD + A′ B′C + A′ BD

3. For an n-variable Boolean function, maximum number of prime implicants

a) 2(n – 1) b) n/2

c) 2^n d) $2^{(n-1)}$

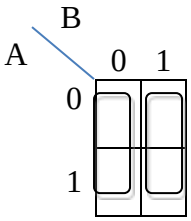
4. Solution:

Prime implicants – Product terms

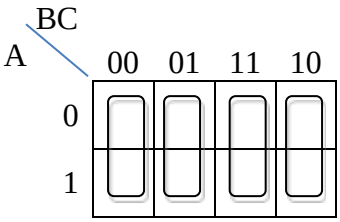
Maximum number of prime implicants in K – Map is obtained when least number of cell combines and least number of cell combinations is obtained by combining two adjacent cell.

∴For 4-variable K – Map

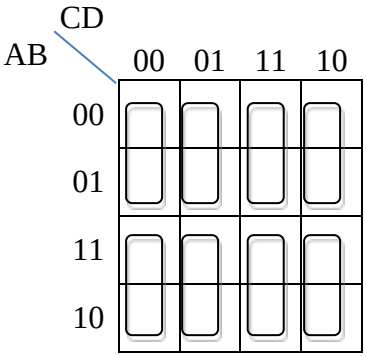
Maximum number of prime implicants



2 Variables



3 Variables



4 Variables

Solution 2^{n-1}

5. The 2 motors M2 & M1 controlled by three sensors S3, S2, S1. One motor M2 is to run any time all three sensors are ON. The other motor is to run whenever sensors S2 or S1 but not both are ON and S3 is OFF. For all sensor combinations where M1 is ON, M2 is to be OFF except when all the three sensors are OFF & then both motors must remain OFF. Construct the truth table and write the Boolean output equation.

S1	S2	S3	M1	M2
0	0	0	0	0
0	0	1	X	X
0	1	0	1	0
0	1	1	X	X
1	0	0	1	0
1	0	1	X	X
1	1	0	X	X
1	1	1	0	1

S1 \ S2 S3	00	01	11	10
0				1
1	1			

S1 \ S2 S3	00	01	11	10
00				
01			1	

M1 = S1 S2′ S3′ + S1′ S2 S3 M2 = S1 S2 S3

- (1987) – The subtraction of a binary number Y from another binary number X, done by adding 2's compliment of Y to X, results in a binary number without overflow, this implies that the result is
  a. Negative and is in normal form
  b. Negative and is in 2's compliment form.
  c. Positive and is in normal form.
  d. Positive and is in 2's compliment form.

Solution:

X, Y.

X – Y = X + (-Y) = X + (2's complement of Y) = SUM

Example: [3 – 5 = -2]

Step 1

```
    011 (X)          011
  – 101 (Y)          011
(2's complement of 101 is 011)   0110 ⟶   6 [2's complement form]
```

Step 2

Take 2's compliment of 110

```
  001
  + 1
  0010 (-2) [Normal form]
```

Carry out '0' indicates its negative.

If sum produces a carry, then dis-carry the carry & the result is positive, which is in normal form. If sum doesn't produce carry, the result is negative and is in 2's compliment form.

- (1993) – 2's complement representation of a 16 bit number (one sign bit and 15 magnitude bits) is FFFF. Its magnitude in decimal representation is,

  a) 0 b) 1

  c) 32,767 d) 65,535

  FFFF ⇒ 1111 1111 1111 1111

  2's compliment of given value is

  0000 0000 0000 0001

  $(0001)_H = (1)_{decimal}$

- (1998) – An equivalent 2's compliment representation of the 2's compliment number 1101 is

  Solution:

  1101 2's positive sign = 0

  MSB Magnitude is 3 negative if sign = 1

  (sign)

  – 3 ⇒ 1 101

  With 4bits representation sign Magnitude

  – 3 with 6 bits representation 111101

  a) 110100

  b) 001101

  c) 110111

  d) 111101

- Two 2's complement numbers having sign bits X and Y are added and the sign bit of the result is Z. Then, the occurrence of overflow is indicated by the Boolean function.

  a) XYZ

  b) X′Y′Z + XYZ′

  c) X′Y′Z′

  d) XY + YZ + XZ

  Range of 2's complement condition to overflow

  -2^{n-1} to $+(2^{n-1} - 1)$

  If n = 4 then, – 8 + 0 + 7

  1) Sum of positive numbers result in a negative number – X = 0, Y = 0, Z = 1

  2) Sum of negative numbers result in a positive number X = 1, Y = 1, Z = 0

  X′Y′Z + XYZ′

- (2014) The number of bytes required to represent the decimal number 1856357 in packed BCD (Binary Coded Decimal form is 4 Byte .

  1 8 5 6 3 5 7

  $\underbrace{00000001}_{\text{Byte 1}}\ \underbrace{10000101}_{\text{Byte 2}}\ \underbrace{01100011}_{\text{Byte 3}}\ \underbrace{01010111}_{\text{Byte4}}$

- (2008) The two numbers represented in signed 2's complement form are P = 11101101 & Q = 11100110. If Q is subtracted from P, the value obtained in signed 2's complement form is,

  Solution

  P = 1 1 1 0 1 1 0 1

  Q = 1 1 1 0 0 1 1 0 → 2's complement 0 0 0 1 1 0 1 0

  1 1 1 0 1 1 0 1
  0 0 0 1 1 0 1 0
  1 0 0 0 0 0 1 1 1 ⇒ **+ 7**

- (2007) X = 0110 & y = 11001 are two 5-bit binary numbers represented in 2's compliment format. The sum of X and Y represented in 2's complement format using 6-bits is.

  0 1 1 1 0
  1 1 0 0 1
  1 0 0 1 1 1

  ⇒ **0 0 0 1 1 1**

- (2005) Decimal 43 in Hexadecimal and BCD number system is respectively.

16	43
	2 ---11

  43 = 0100 0011

  2, B = 11,

  2B, 0100 0011

- (2004) The range of signed decimal numbers that can be represented by 6-bit 1's complement number is

  $(-2^{n-1} - 1)$ to $(+2^{n-1} - 1)$

  $(-2^{5-1} - 1)$ to $(+2^{5-1} - 1)$

  – 31 to + 31

- (2002) 4-bit 2's complement representation of a decimal number is 1000. The number is

  1000 → (– 8)
  0111
  + 1 .
  01000 ⇒ (+8)

- (2001) The 2's complement representation of –17 is

  $17 = 2^4 + 2^0 = 16 + 1$

  010001 2's complement 101110
  +1
  101111

- (2004) A new BCD number system is proposed in which every digit of a base – 5 number is represented by its corresponding 3–bit binary code. For example the base-5 number 24 will be represented by its BCP code 010100. In this numbering system, the BCP code 100010011001 corresponds to the following number in base-5 system.

  2 4 BCP code
  010 100 → 010 100
  100 010 011 001
  4 2 3 1

- (2004) 11001, 1001 & 111001 correspond to the 2's complement representation of which one of the following sets of number?

  11001 → 00110
  + 1
  00111 → 7

  1001 → 0110
  +1
  0111 → 7

  111001 → 000110
  + 1
  000111 → 7

- (2015) A function of Boolean variables X, Y & Z is expressed in terms of the min terms as F(X, Y, Z) = Σ(1, 2, 5, 6. 7) which one of the product of sums given below is equal to the function F(X, Y, Z)?

  a) (X′ + Y′ + Z′) (X′ + Y + Z) (X + Y′ + Z′)

  b) (X + Y + Z) (X + Y′ + Z′) (X′ + Y + Z)

  c) (X′ + Y′ + Z) (X′ + Y + Z′) (X + Y′ + Z) (X + Y + Z′) (X + Y + Z)

  d) (X + Y + Z′) (X′ + Y + Z) (X′ + Y + Z′) (X′ + Y′ + Z) (X′ + Y′ + Z′)

  F(X, Y, Z) = Σ(1, 2, 5, 6, 7)

  3 variables SOP

  SOP ⇒ F = 1 if XYZ = 001, 010, 101, 110, 111

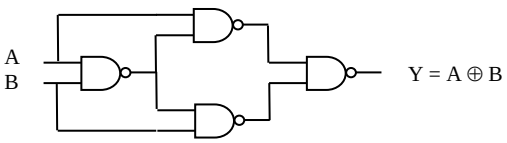
  POS ⇒ F = 0 if XYZ = 000, 011, 100

  F = (X + Y + Z) (X + Y′ + Z′) (X′ + Y + Z)

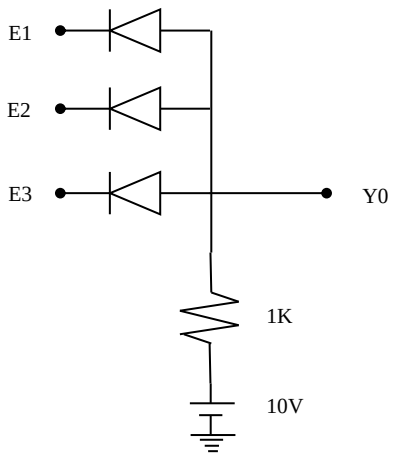
- (2016) The minimum number of 2-input NAND gates required to implement a 2-input XOR gate is, **4**

  Let A & B are two input

  Y = A ⊕ B



- (2015) In the circuit shown, diodes D1, D2 and D3 are ideal and the inputs E1, E2 and E3 are ‘0V’ for logic ‘0’ and ‘10V’ for logic ‘1’, which logic gate does the circuit represent?



If any input is logic ‘0’ then corresponding diode will be ON and output Y0 = 0

If all inputs are logic ‘1’ (10V) then all diodes are OFF and Y0 = 10V.

Y0 = E1 . E2 . E3

AND gate (3 input AND gate)

- (2014) The Boolean expression $(X + Y)(X + Y') + ((XY') + X')'$ simplifies to,

$$
\begin{aligned}
F &= (X + Y)(X + Y') + ((XY') + X')' \\
&= X X + X Y' + X Y + Y Y' + ((X + X')(X' + Y'))' \\
&= X + XY' + XY + ((X + X')(X' + Y'))' \\
&= X(1 + Y') + XY + (X' + Y')' \\
&= X(1 + Y') + XY + XY \\
&= X + XY + XY \\
&= X(1 + Y) + XY \\
&= X + XY \\
&= X(1 + Y') \\
&= X
\end{aligned}
$$

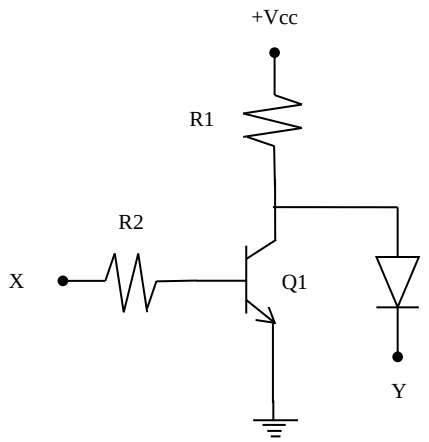
- (2013) A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by any one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles.

Switch		**State of Bulb**
A	**B**	**Y**
0	0	0
0	1	1
1	0	1
1	1	0

$Y = A'B + AB'$

XOR gate

- (2013)



In the circuit below Q1 has negligible collector to emitter saturation voltage and the diode drops negligible voltage across it under forward bias. If Vcc is +5V, X and Y are digital signals with 0V as logic ‘0’ and Vcc as logic ‘1’, then Boolean expression for Z is.

X	Y	Q	Diode Z
0	0	OFF	Forward 0
0	1	OFF	Reverse 1
1	0	ON	Forward 0
1	1	ON	Reverse 0

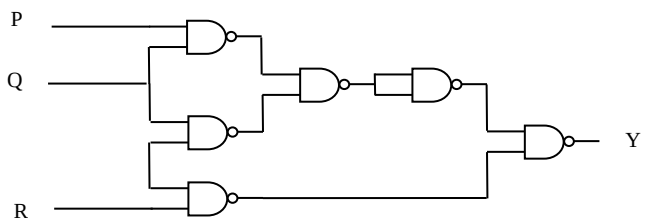
$$Z = X'Y$$

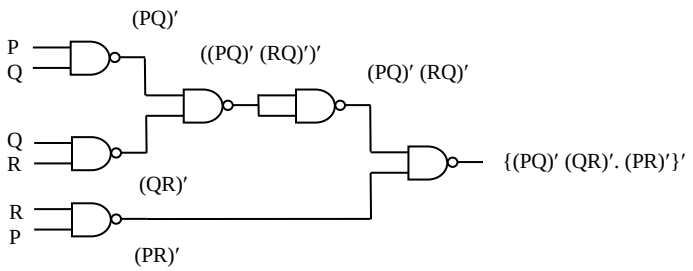
❖ (2012) In the Sum of products functions F(X, Y, Z) = Σ(2, 3, 4,5) the prime implicants are

X \ YZ	00	01	11	10
0	0	0	1	1
1	1	1	0	0

$\Rightarrow X'Y, XY'$

❖ (2012) The output Y in the circuit below is always ‘1’ when



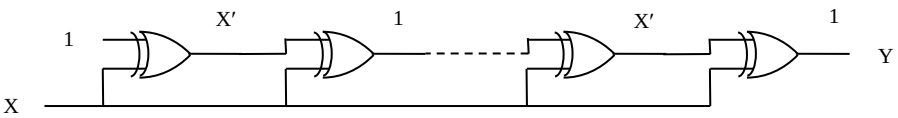


$$Y = ((PQ)'(QR)'(PR)')'$$
$$= PQ + QR + PR$$

Two or more of the inputs, P, Q, R are ‘1’

- When number of distinct Boolean expression of 4 variables is
  $2^{2(n)} = 2^{2(4)} = 2^{2(16)} = 65{,}536$

- If the input to the digital circuit shown in the figure, consisting of cascade of 20 XOR gates is X then the output Y is equal to



  Even n% of XOR gates result in 1,
  So 20 number of XOR → Y = 1

- (2017) Which one of the following gives the simplified sum of products expression for the Boolean function $F = m_0 + m_2 + m_3 + m_5$, where m_0, m_2, m_3, m_5 are minterms corresponding to the inputs A, B & C with A is the MSB and C as the LSB.

x \ yz	00	01	11	10
0	m0	m1	m3	m2
1	m4	m5	m7	m6

  $F = (m_0 + m_2) + (m_2 + m_3) + m_5$
  $= A'C' + A'B + AB'C$

- (2015) a 3-input majority gate is defined by the logic function M(a, b, c) = ab + bc + ca. Which one of the following gates is represented by the function.

  M [(M(a, b, c))′, (M(a, b, c′),c)] ?

  M(a, b, c) = ab + bc + ca
  (M(a, b, c))′ = (ab + bc + ca)′
  = (ab)′ . (bc)′ . (ca)′
  = (a′ + b′) (b′ + c′) (c′ + a′)
  = (ab)′ + (bc)′ + (ca)′
  M(a, b, c′) = (ab) + (bc′) + (c′a)

$$M[(M(a, b, c))', (M(a, b, c'), c)] = ((ab)' + (bc)' + (ca)') + ((ab) + (bc') + (c'a))$$
$$= [((ab)' + (bc)' + (ca)') + ((ab) + (bc') + (c'a))c]$$
$$= (a'bc') + (ab'c') + (abc) + (a'b'c)$$
$$= \Sigma m(1, 2, 4, 7)$$
$$= a \oplus b \oplus c$$

3 input XOR gate.